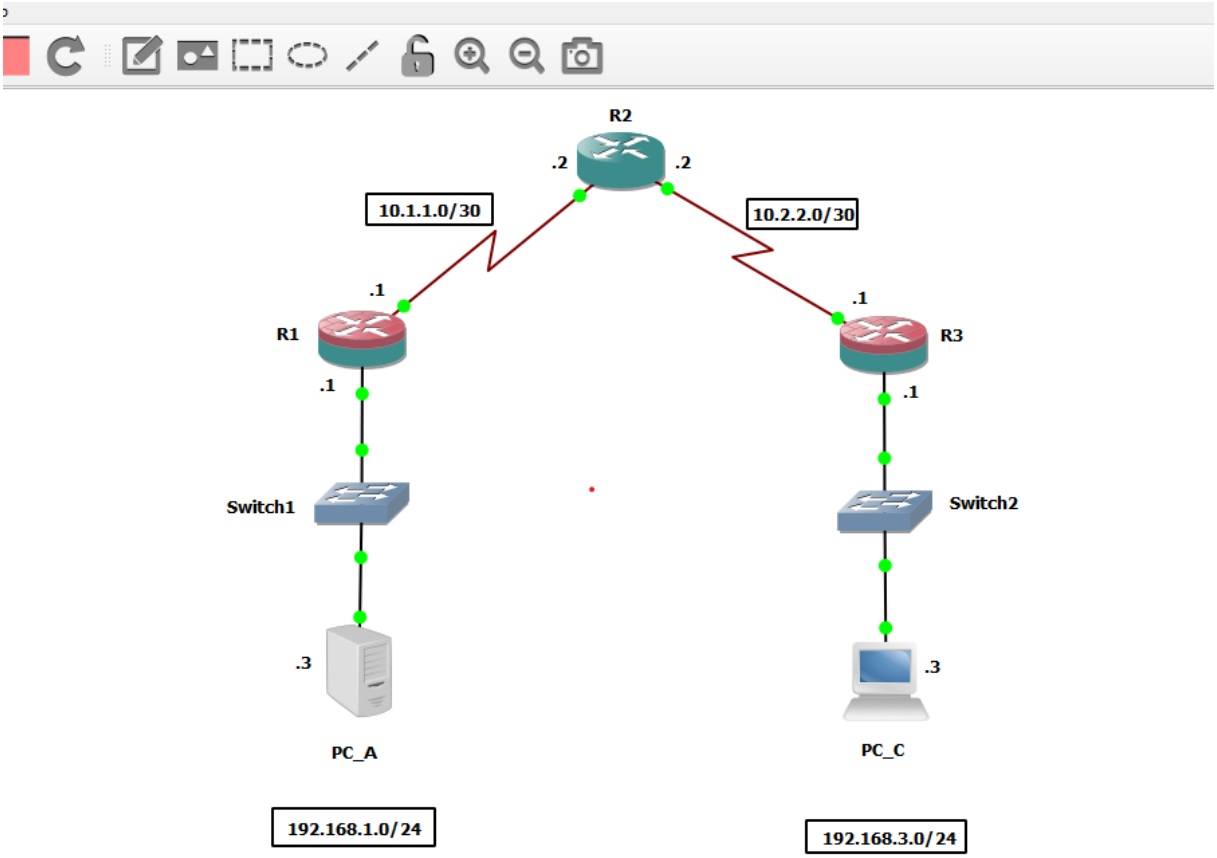
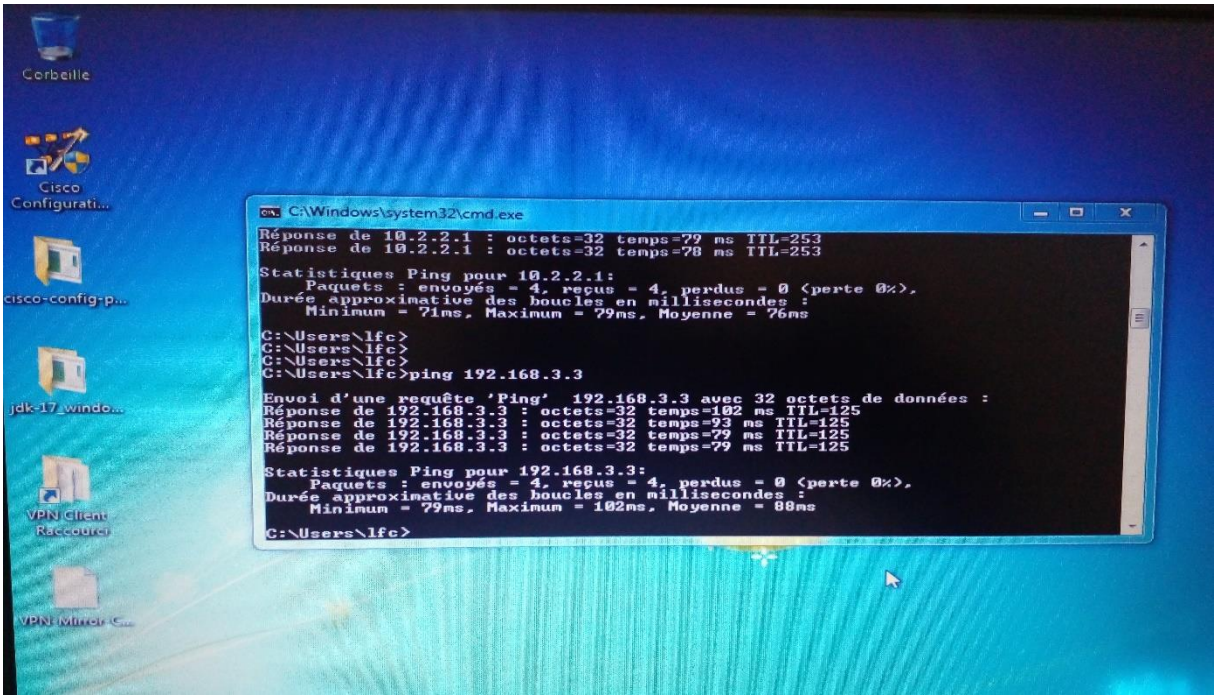
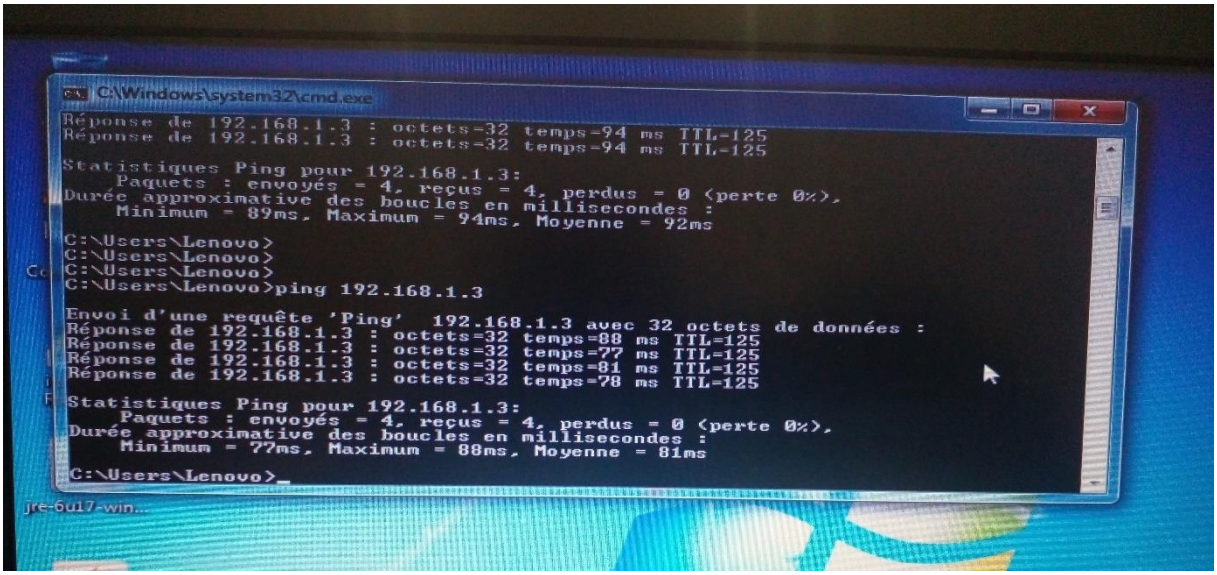


- **<u>Partie I :</u> Configuration de base**





- **<u>Partie II :</u> Configuration Site to Site VPN (CLI)**

R1 R2 R3

```
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/43/56 ms
R1#
R1#
R1#
R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#
R1(config)#cry
R1(config)#crypto is
R1(config)#crypto isakmp en
R1(config)#crypto isakmp enable
R1(config)#
R1(config)#
R1(config)#
R1(config)#
R1(config)#crypto is
R1(config)#crypto isakmp pl
R1(config)#crypto isakmp p
R1(config)#crypto isakmp po
R1(config)#crypto isakmp policy 1C
                                 ^
% Invalid input detected at '^' marker.

R1(config)#crypto isakmp policy 10
R1(config-isakmp)#
R1(config-isakmp)#
R1(config-isakmp)#au
R1(config-isakmp)#authentication pre
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#en
R1(config-isakmp)#encryption ae
R1(config-isakmp)#encryption aes 2(
R1(config-isakmp)#encryption aes 256
R1(config-isakmp)#
R1(config-isakmp)#has
R1(config-isakmp)#hash sha
R1(config-isakmp)#group 5
R1(config-isakmp)#lif
R1(config-isakmp)#lifetime 3600
R1(config-isakmp)#end
R1#
R1#
R1#
*Dec 20 12:42:52.779: %SYS-5-CONFIG_I: Configured from console by console
R1#
R1#
R1#
```

R1 R2 R3

```
R3(config)#crypto isakmp policy 10
R3(config-isakmp)#
R3(config-isakmp)#au
R3(config-isakmp)#authentication pre
R3(config-isakmp)#authentication pre-share
R3(config-isakmp)#
R3(config-isakmp)#en
R3(config-isakmp)#encryption aes
R3(config-isakmp)#encryption aes 256
R3(config-isakmp)#
R3(config-isakmp)#
R3(config-isakmp)#has
R3(config-isakmp)#hash sh
R3(config-isakmp)#hash sha
R3(config-isakmp)#
R3(config-isakmp)#gro
R3(config-isakmp)#group 5
R3(config-isakmp)#
R3(config-isakmp)#
R3(config-isakmp)#li
R3(config-isakmp)#lifetime 3600
R3(config-isakmp)#
R3(config-isakmp)#
R3(config-isakmp)#end
R3#
*Dec 20 12:47:22.015: %SYS-5-CONFIG_I: Configured from console by console
R3#
R3#
R3#
R3#sh cry
R3#sh crypto is
R3#sh crypto isakmp po
R3#sh crypto isakmp policy

Global IKE policy
Protection suite of priority 10
        encryption algorithm:   AES - Advanced Encryption Standard (256 bit keys).
        hash algorithm:         Secure Hash Standard
        authentication method:  Pre-Shared Key
        Diffie-Hellman group:   #5 (1536 bit)
        lifetime:               3600 seconds, no volume limit
R3#
R3#
R3#
R3#
R3#
R3#
R3#
R3#
R3#
R3#
```

R1 R2 R3

```
R1(config-crypto-map)#set
R1(config-crypto-map)#set tr
R1(config-crypto-map)#set transform-set 50
R1(config-crypto-map)#set sec
R1(config-crypto-map)#set security-association li
R1(config-crypto-map)#set security-association lifetime sec
R1(config-crypto-map)#set security-association lifetime seconds 900
R1(config-crypto-map)#exit
R1(config)#cruop
R1(config)#cryp
R1(config)#crypto ma
R1(config)#crypto map CMAP 10
R1(config-crypto-map)#set pfs grou
R1(config-crypto-map)#set pfs group5
R1(config-crypto-map)#
R1(config-crypto-map)#
R1(config-crypto-map)#
R1(config-crypto-map)#exit
R1(config)#
R1(config)#
R1(config)#
R1(config)#
R1(config)#
R1(config)#in
R1(config)#interface s3/0
R1(config-if)#cry
R1(config-if)#crypto ma
R1(config-if)#crypto map CMAP
R1(config-if)#
R1(config-if)#
*Dec 20 13:19:40.871: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R1(config-if)#end
R1#
R1#
R1#
*Dec 20 13:21:15.447: %SYS-5-CONFIG_I: Configured from console by console
R1#
R1#
R1#sh cr
R1#sh cry
R1#sh crypto ip
R1#sh crypto ipsec r
R1#sh crypto ipsec tr
R1#sh crypto ipsec transform-set
Transform set default: { esp-aes esp-sha-hmac  }
   will negotiate = { Transport,  },

Transform set 50: { esp-256-aes esp-sha-hmac  }
   will negotiate = { Tunnel,  },

R1#
```

R1 R2 R3

```
R3(config-if)#
R3(config-if)#cry
R3(config-if)#crypto ma
R3(config-if)#crypto map CMAP
R3(config-if)#
*Dec 20 13:19:57.663: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R3(config-if)#
R3(config-if)#
R3(config-if)#end
R3#
R3#
R3#
*Dec 20 13:20:03.599: %SYS-5-CONFIG_I: Configured from console by console
R3#
R3#
R3#sh cry
R3#sh crypto is
R3#sh crypto ip
R3#sh crypto ipsec tr
R3#sh crypto ipsec transform-set
Transform set default: { esp-aes esp-sha-hmac  }
   will negotiate = { Transport,  },

Transform set 50: { esp-256-aes esp-sha-hmac  }
   will negotiate = { Tunnel,  },

R3#
R3#
R3#
R3#sh cr
R3#sh cry
R3#sh crypto ma
R3#sh crypto map
Crypto Map IPv4 "CMAP" 10 ipsec-isakmp
        Peer = 10.1.1.1
        Extended IP access list 101
            access-list 101 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
        Current peer: 10.1.1.1
        Security association lifetime: 4608000 kilobytes/900 seconds
        Responder-Only (Y/N): N
        PFS (Y/N): Y
        DH group:  group5
        Mixed-mode : Disabled
        Transform sets={
                50:  { esp-256-aes esp-sha-hmac  } ,
        }
        Interfaces using crypto map CMAP:
                Serial3/0


R3#
```

R1 R2 R3

```
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#sh cryo
R1#sh cry
R1#sh crypto isa
R1#sh crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id status

IPv6 Crypto ISAKMP SA

R1#sh crypto ipsec sa

interface: Serial3/0
    Crypto map tag: CMAP, local addr 10.1.1.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
   remote ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
   current_peer 10.2.2.1 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
    #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

     local crypto endpt.: 10.1.1.1, remote crypto endpt.: 10.2.2.1
     plaintext mtu 1500, path mtu 1500, ip mtu 1500, ip mtu idb Serial3/0
     current outbound spi: 0x0(0)
     PFS (Y/N): N, DH group: none

     inbound esp sas:

     inbound ah sas:

     inbound pcp sas:

     outbound esp sas:

     outbound ah sas:

     outbound pcp sas:
R1#
R1#
```

R1 R2 R3

```
R3#
R3#sh cry
R3#sh crypto isa
R3#sh crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id status

IPv6 Crypto ISAKMP SA

R3#sh cry
R3#sh crypto ip
R3#sh crypto ipsec sa

interface: Serial3/0
    Crypto map tag: CMAP, local addr 10.2.2.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
   remote ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
   current_peer 10.1.1.1 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
    #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

     local crypto endpt.: 10.2.2.1, remote crypto endpt.: 10.1.1.1
     plaintext mtu 1500, path mtu 1500, ip mtu 1500, ip mtu idb Serial3/0
     current outbound spi: 0x0(0)
     PFS (Y/N): N, DH group: none

     inbound esp sas:

     inbound ah sas:

     inbound pcp sas:

     outbound esp sas:

     outbound ah sas:

     outbound pcp sas:
R3#
R3#
R3#
R3#
R3#
R3#
R3#
```

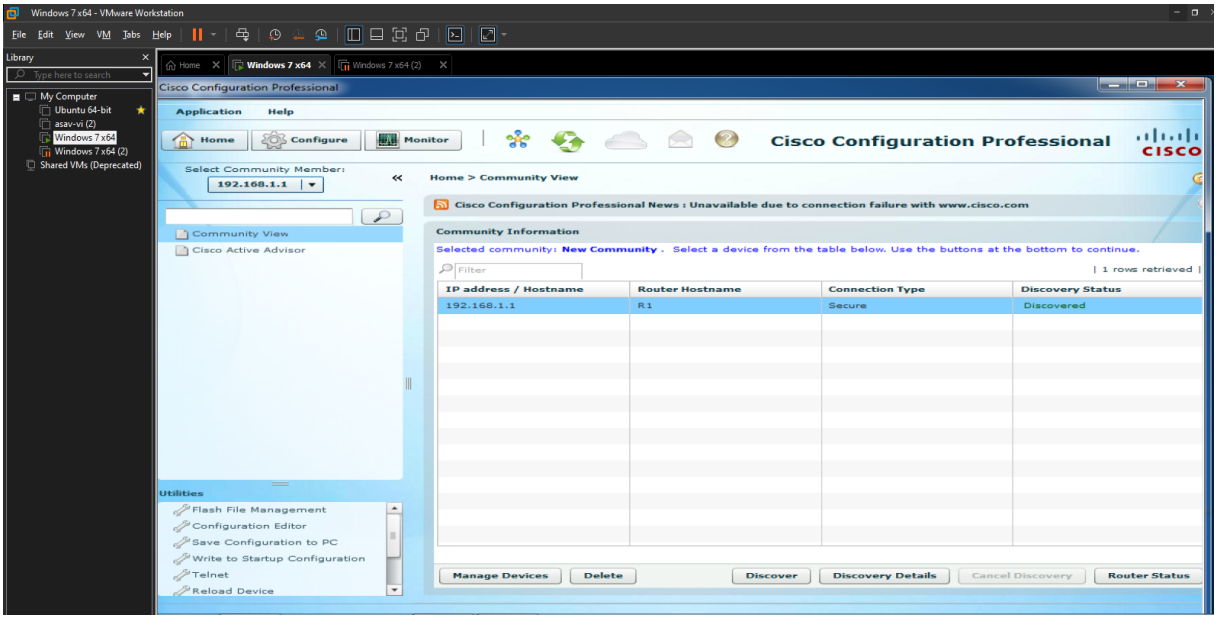
R1 R2 R3

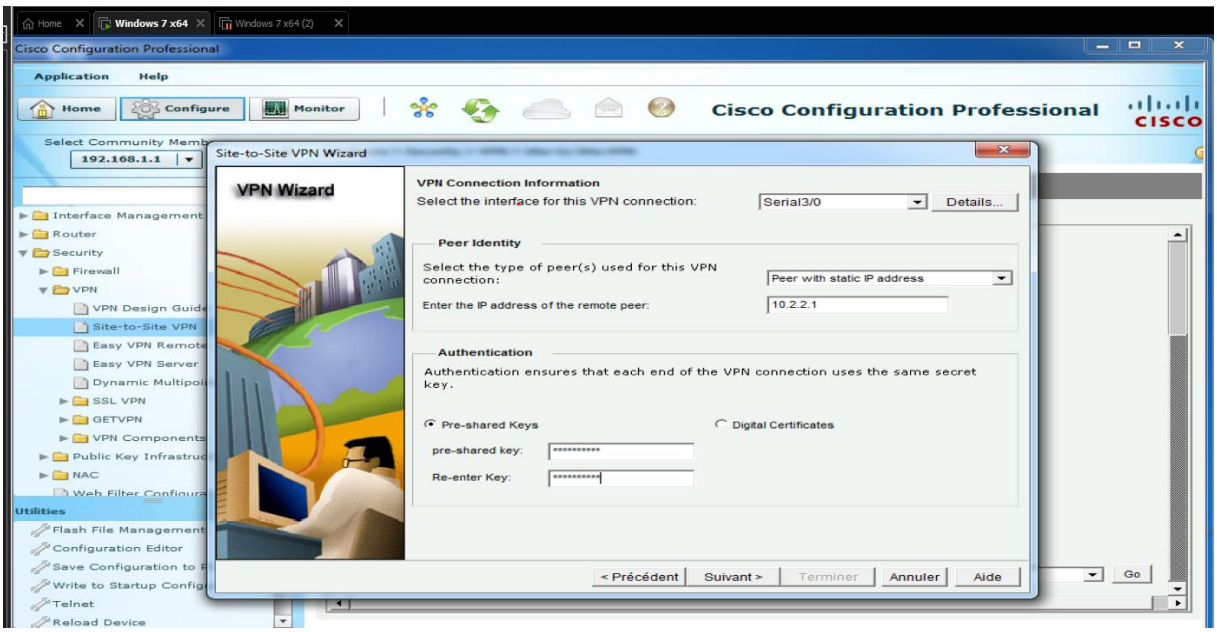
```
R1(config)#
R1(config)#deb
R1(config)#debu
R1(config)#debu
R1(config)#debu
R1(config)#exit
R1#
R1#
R1#deb
R1#debug
*Dec 20 13:32:38.331: %SYS-5-CONFIG_I: Configured from console by console
R1#debug ip os
R1#debug ip ospf hello
OSPF hello debugging is on
R1#
*Dec 20 13:32:46.247: OSPF-101 HELLO Fa0/0: Send hello to 224.0.0.5 area 0 from 192.168.1.1
R1#
*Dec 20 13:32:47.499: OSPF-101 HELLO Se3/0: Send hello to 224.0.0.5 area 0 from 10.1.1.1
R1#
*Dec 20 13:32:55.443: OSPF-101 HELLO Fa0/0: Send hello to 224.0.0.5 area 0 from 192.168.1.1
R1#
*Dec 20 13:32:57.303: OSPF-101 HELLO Se3/0: Send hello to 224.0.0.5 area 0 from 10.1.1.1
R1#
*Dec 20 13:33:05.423: OSPF-101 HELLO Fa0/0: Send hello to 224.0.0.5 area 0 from 192.168.1.1
R1#
*Dec 20 13:33:06.867: OSPF-101 HELLO Se3/0: Send hello to 224.0.0.5 area 0 from 10.1.1.1
R1#
*Dec 20 13:33:08.583: OSPF-101 HELLO Se3/0: Rcv hello from 10.2.2.2 area 0 10.1.1.2
R1#
*Dec 20 13:33:14.447: OSPF-101 HELLO Fa0/0: Send hello to 224.0.0.5 area 0 from 192.168.1.1
R1#
*Dec 20 13:33:16.859: OSPF-101 HELLO Se3/0: Send hello to 224.0.0.5 area 0 from 10.1.1.1
R1#
*Dec 20 13:33:24.079: OSPF-101 HELLO Fa0/0: Send hello to 224.0.0.5 area 0 from 192.168.1.1
R1#
*Dec 20 13:33:26.059: OSPF-101 HELLO Se3/0: Send hello to 224.0.0.5 area 0 from 10.1.1.1
R1#
*Dec 20 13:33:33.303: OSPF-101 HELLO Se3/0: Rcv hello from 10.2.2.2 area 0 10.1.1.2
*Dec 20 13:33:33.635: OSPF-101 HELLO Fa0/0: Send hello to 224.0.0.5 area 0 from 192.168.1.1
R1#
*Dec 20 13:33:35.735: OSPF-101 HELLO Se3/0: Send hello to 224.0.0.5 area 0 from 10.1.1.1
R1#
*Dec 20 13:33:43.555: OSPF-101 HELLO Fa0/0: Send hello to 224.0.0.5 area 0 from 192.168.1.1
R1#
*Dec 20 13:33:45.127: OSPF-101 HELLO Se3/0: Send hello to 224.0.0.5 area 0 from 10.1.1.1
R1#
*Dec 20 13:33:53.167: OSPF-101 HELLO Se3/0: Rcv hello from 10.2.2.2 area 0 10.1.1.2
*Dec 20 13:33:53.483: OSPF-101 HELLO Fa0/0: Send hello to 224.0.0.5 area 0 from 192.168.1.1
R1#
*Dec 20 13:33:54.351: OSPF-101 HELLO Se3/0: Send hello to 224.0.0.5 area 0 from 10.1.1.1
R1#
```

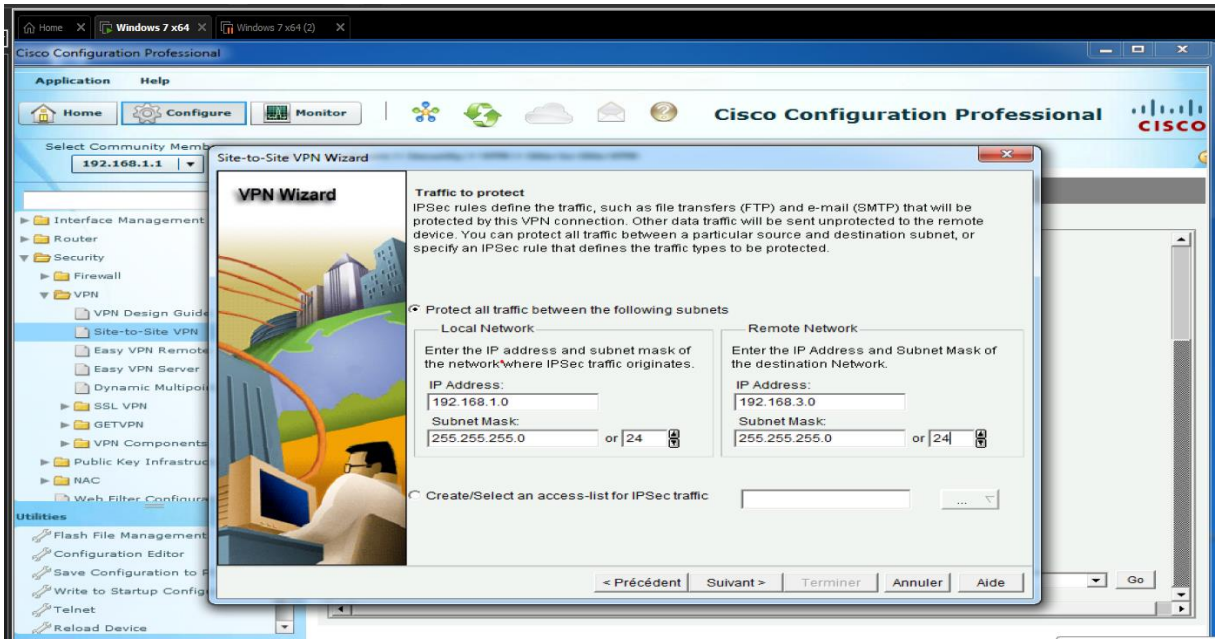
R1 R2 R3

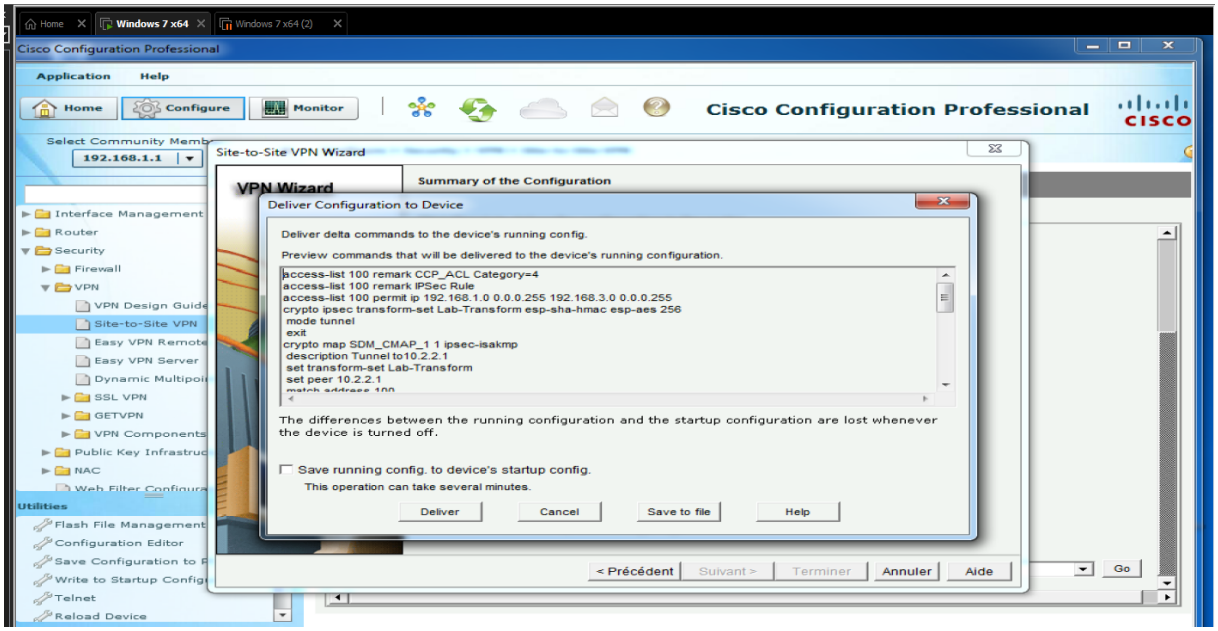
```
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#ping
Protocol [ip]:
Target IP address: 192.168.3.1
Repeat count [5]:
Datagram size [100]:
Timeout in seconds [2]:
Extended commands [n]: y
Source address or interface: 192.168.1.1
Type of service [0]:
Set DF bit in IP header? [no]:
Validate reply data? [no]:
Data pattern [0xABCD]:
Loose, Strict, Record, Timestamp, Verbose[none]:
Sweep range of sizes [n]:
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
..!!!
Success rate is 60 percent (3/5), round-trip min/avg/max = 80/90/96 ms
R1#
R1#
R1#
R1#sh crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id status
10.2.2.1        10.1.1.1        QM_IDLE           1001 ACTIVE

IPv6 Crypto ISAKMP SA

R1#
R1#
R1#
```

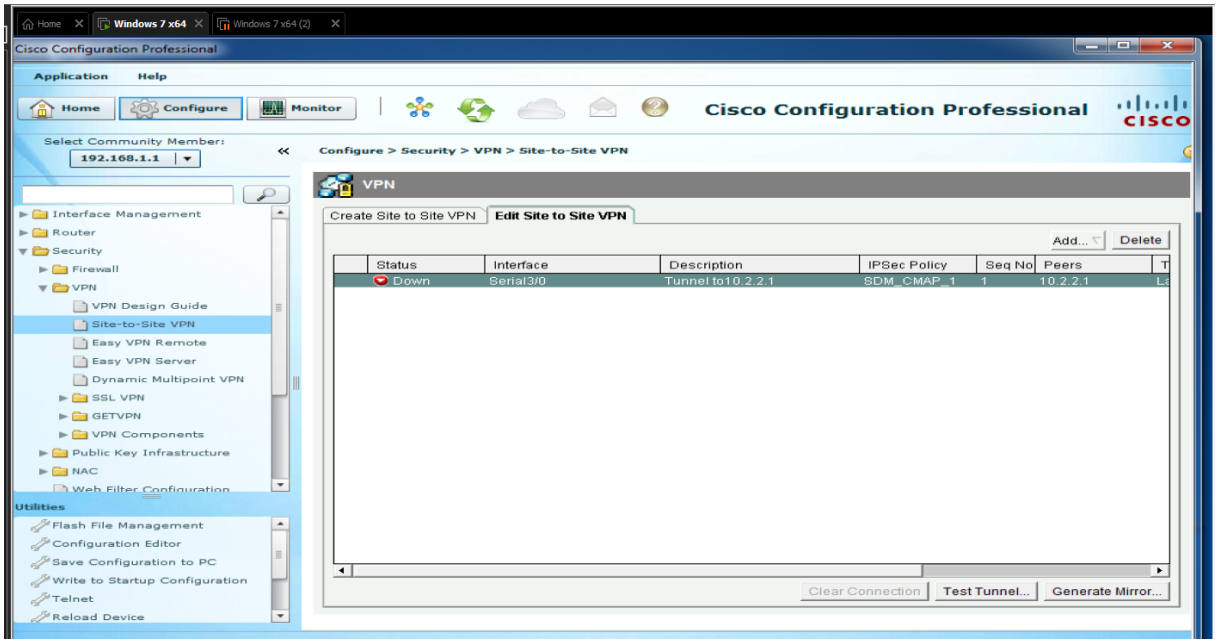
➢ <u>Partie III :</u> Configuration Site to Site VPN (CCP)

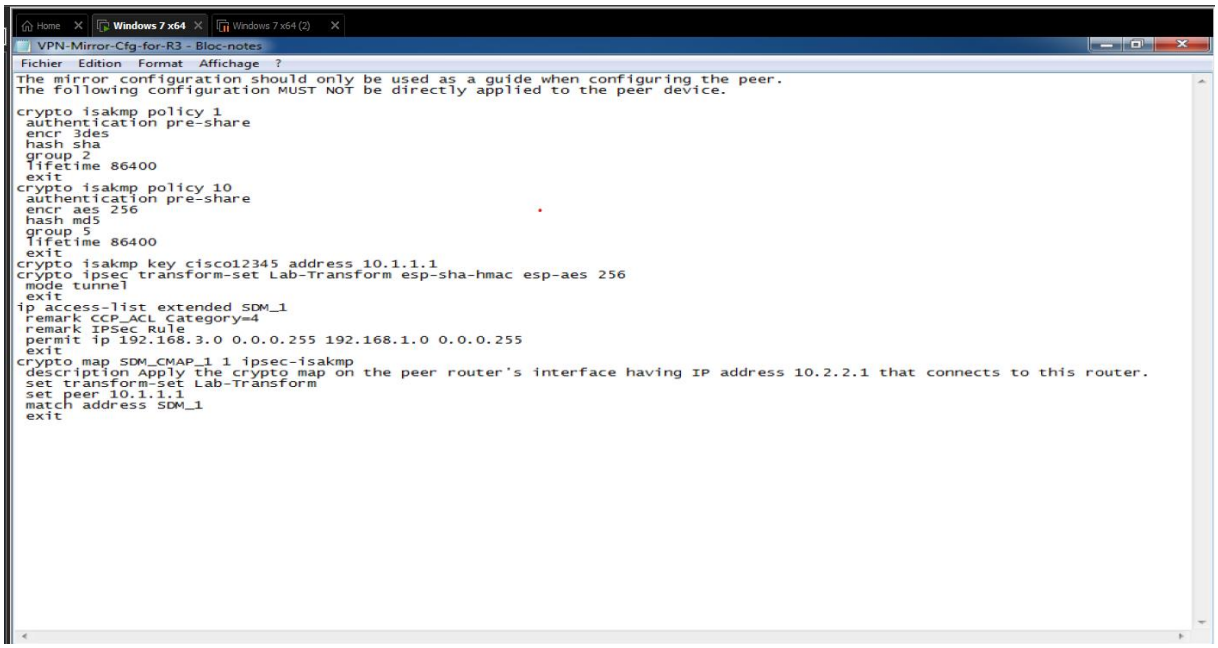





Cisco Configuration Professional
Application Help
Home Configure Monitor
Cisco Configuration Professional
Site-to-Site VPN Wizard
VPN Wizard
Traffic to protect
IPSec rules define the traffic, such as file transfers (FTP) and e-mail (SMTP) that will be protected by this VPN connection. Other data traffic will be sent unprotected to the remote device. You can protect all traffic between a particular source and destination subnet, or specify an IPSec rule that defines the traffic types to be protected.
Protect all traffic between the following subnets
Local Network
Enter the IP address and subnet mask of the network where IPSec traffic originates.
IP Address:
192.168.1.0
Subnet Mask:
255.255.255.0
or 24
Remote Network
Enter the IP Address and Subnet Mask of the destination Network.
IP Address:
192.168.3.0
Subnet Mask:
255.255.255.0
or 24
Create/Select an access-list for IPSec traffic
< Précédent
Suivant >
Terminer
Annuler
Aide


Cisco Configuration Professional
Application Help
Home Configure Monitor
Cisco Configuration Professional
Site-to-Site VPN Wizard
VPN Wizard
Summary of the Configuration
Deliver Configuration to Device
Deliver delta commands to the device's running config.
Preview commands that will be delivered to the device's running configuration.
access-list 100 remark CCP_ACL Category=4
access-list 100 remark IPSec Rule
access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
crypto ipsec transform-set Lab-Transform esp-sha-hmac esp-aes 256
mode tunnel
exit
crypto map SDM_CMAP_1 1 ipsec-isakmp
description Tunnel to10.2.2.1
set transform-set Lab-Transform
set peer 10.2.2.1
The differences between the running configuration and the startup configuration are lost whenever the device is turned off.
Save running config. to device's startup config.
This operation can take several minutes.
Deliver
Cancel
Save to file
Help
< Précédent
Suivant >
Terminer
Annuler
Aide


Home
Windows 7 x64
Windows 7 x64 (2)
Cisco Configuration Professional
Application
Help
Home
Configure
Monitor
Cisco Configuration Professional
CISCO
Select Community Member:
192.168.1.1
Configure > Security > VPN > Site-to-Site VPN
Interface Management
Router
Security
Firewall
VPN
VPN Design Guide
Site-to-Site VPN
Easy VPN Remote
Easy VPN Server
Dynamic Multipoint VPN
SSL VPN
GETVPN
VPN Components
Public Key Infrastructure
NAC
Web Filter Configuration
Utilities
Flash File Management
Configuration Editor
Save Configuration to PC
Write to Startup Configuration
Telnet
Reload Device
VPN
Create Site to Site VPN
Edit Site to Site VPN
Add...
Delete
Status
Interface
Description
IPSec Policy
Seq No
Peers
Down
Serial3/0
Tunnel to10.2.2.1
SDM_CMAP_1
1
10.2.2.1
Clear Connection
Test Tunnel...
Generate Mirror...


Home
Windows 7 x64
Windows 7 x64 (2)
VPN-Mirror-Cfg-for-R3 - Bloc-notes
Fichier
Edition
Format
Affichage
?
The mirror configuration should only be used as a guide when configuring the peer.
The following configuration MUST NOT be directly applied to the peer device.
crypto isakmp policy 1
authentication pre-share
encr 3des
hash sha
group 2
lifetime 86400
exit
crypto isakmp policy 10
authentication pre-share
encr aes 256
hash md5
group 5
lifetime 86400
exit
crypto isakmp key cisco12345 address 10.1.1.1
crypto ipsec transform-set Lab-Transform esp-sha-hmac esp-aes 256
mode tunnel
exit
ip access-list extended SDM_1
remark CCP_ACL Category=4
remark IPSec Rule
permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
exit
crypto map SDM_CMAP_1 1 ipsec-isakmp
description Apply the crypto map on the peer router's interface having IP address 10.2.2.1 that connects to this router.
set transform-set Lab-Transform
set peer 10.1.1.1
match address SDM_1
exit

R1 R2 R3

```
!
!
!
gatekeeper
 shutdown
!
!
line con 0
 exec-timeout 5 0
 privilege level 15
 password 7 060506324F410A160B0713181F
 logging synchronous
 login
 stopbits 1

R1#
R1#
R1#
R1#
R1#tel
R1#telnet 10.2.2.1
Trying 10.2.2.1 ... Open

[Connection to 10.2.2.1 closed by foreign host]
R1#telnet 10.2.2.1
Trying 10.2.2.1 ... Open


User Access Verification

Password:
Password:
Password:
% Bad passwords

[Connection to 10.2.2.1 closed by foreign host]
R1#telnet 10.2.2.1
Trying 10.2.2.1 ... Open


User Access Verification

Password:
R3>
R3>
R3>en
Password:
R3#
R3#
R3#
R3#
```

R1 R2 R3

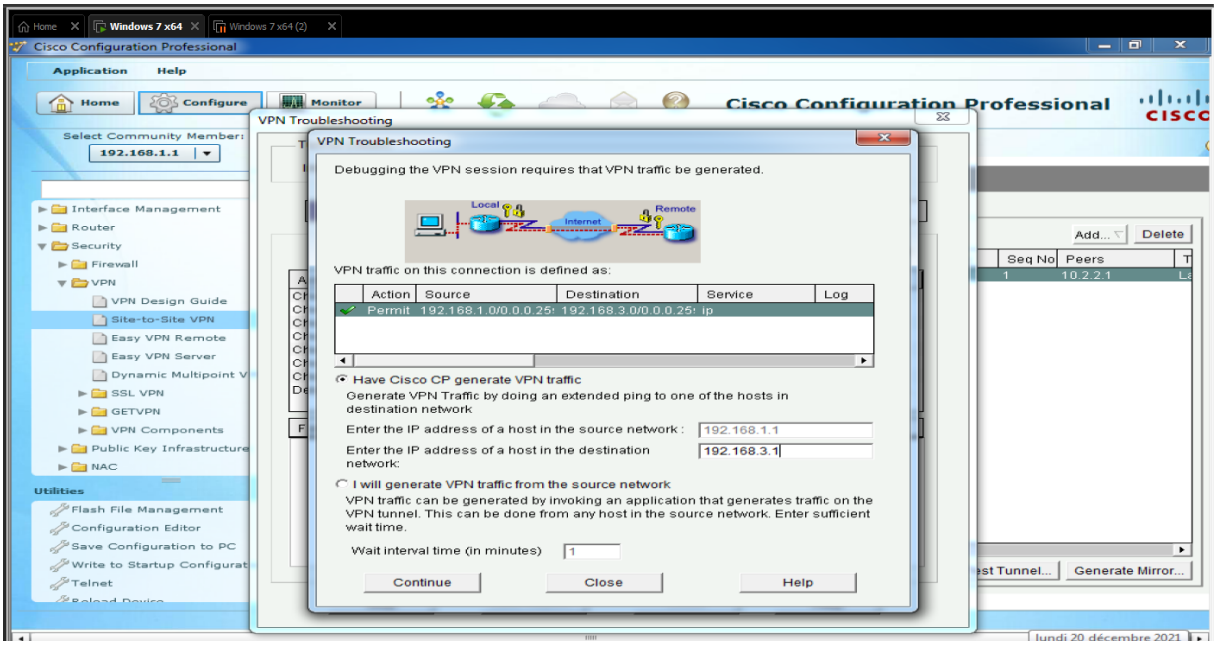
```
R3(config-if)#
R3(config-if)#
R3(config-if)#cry
R3(config-if)#crypto ma
R3(config-if)#crypto map SDM_CMAP_1
ERROR: Crypto Map with tag "SDM_CMAP_1" does not exist.

R3(config-if)#exit
R3(config)#
R3(config)#
R3(config)#
R3(config)#
R3(config)#crypto isakmp policy 1
R3(config-isakmp)# authentication pre-share
R3(config-isakmp)# encr 3des
R3(config-isakmp)# hash sha
R3(config-isakmp)# group 2
R3(config-isakmp)# lifetime 86400
R3(config-isakmp)# exit
R3(config)#crypto isakmp policy 10
R3(config-isakmp)# authentication pre-share
R3(config-isakmp)# encr aes 256
R3(config-isakmp)# hash md5
R3(config-isakmp)# group 5
R3(config-isakmp)# lifetime 86400
R3(config-isakmp)# exit
R3(config)#crypto isakmp key cisco12345 address 10.1.1.1
R3(config)#crypto ipsec transform-set Lab-Transform esp-sha-hmac esp-aes 256
R3(cfg-crypto-trans)# mode tunnel
R3(cfg-crypto-trans)# exit
R3(config)#ip access-list extended SDM_1
R3(config-ext-nacl)# remark CCP_ACL Category=4
R3(config-ext-nacl)# remark IPSec Rule
R3(config-ext-nacl)# permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
R3(config-ext-nacl)# exit
R3(config)#crypto map SDM_CMAP_1 1 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R3(config-crypto-map)#$ IP address 10.2.2.1 that connects to this router.
R3(config-crypto-map)# set transform-set Lab-Transform
R3(config-crypto-map)# set peer 10.1.1.1
R3(config-crypto-map)# match address SDM_1
R3(config-crypto-map)# exit
R3(config)#
R3(config)#
R3(config)#
R3(config)#
R3(config)#int s3/0
R3(config-if)#crypto map SDM_CMAP_1
R3(config-if)#
R3(config-if)#
```

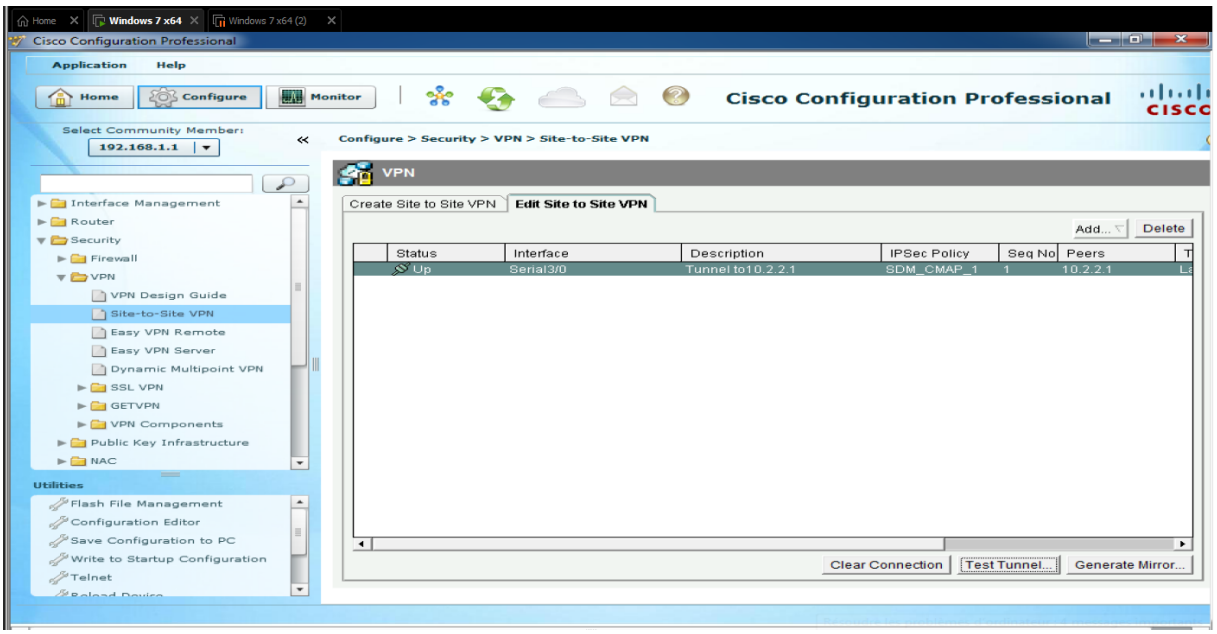
R1 R2 R3

```
R3 con0 is now available

Press RETURN to get started.

*Dec 20 13:59:04.019: %SYS-5-CONFIG_I: Configured from console by console
*Dec 20 14:09:08.019: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
```

R1 R2 R3

```
R3#sh run | begin 3/0
interface Serial3/0
 ip address 10.2.2.1 255.255.255.252
 serial restart-delay 0
 crypto map SDM_CMAP_1
!
interface Serial3/1
 no ip address
 shutdown
 serial restart-delay 0
!
interface Serial3/2
 no ip address
 shutdown
 serial restart-delay 0
!
interface Serial3/3
 no ip address
 shutdown
 serial restart-delay 0
!
interface GigabitEthernet4/0
 no ip address
 shutdown
 negotiation auto
!
interface Ethernet5/0
 no ip address
 shutdown
 duplex half
!
interface Ethernet5/1
 no ip address
 shutdown
 duplex half
!

R3#
R3#
R3#
R3#sh cry
R3#sh crypto ip
R3#sh crypto ipsec tr
R3#sh crypto ipsec transform-set
Transform set default: { esp-aes esp-sha-hmac  }
   will negotiate = { Transport,  },

Transform set Lab-Transform: { esp-256-aes esp-sha-hmac  }
   will negotiate = { Tunnel,  },

R3#
```


Cisco Configuration Professional
Application
Help
Home
Configure
Monitor
Select Community Member:
192.168.1.1
Interface Management
Router
Security
Firewall
VPN
VPN Design Guide
Site-to-Site VPN
Easy VPN Remote
Easy VPN Server
SSL VPN
GETVPN
VPN Components
Public Key Infrastructure
NAC
Utilities
Flash File Management
Configuration Editor
Save Configuration to PC
Telnet
VPN Troubleshooting
Debugging the VPN session requires that VPN traffic be generated.
Local
Internet
Remote
VPN traffic on this connection is defined as:
Action
Source
Destination
Service
Log
Permit 192.168.1.0/0.0.0.25 192.168.3.0/0.0.0.25 ip
Have Cisco CP generate VPN traffic
Generate VPN Traffic by doing an extended ping to one of the hosts in destination network
Enter the IP address of a host in the source network :
192.168.1.1
Enter the IP address of a host in the destination network:
192.168.3.1
I will generate VPN traffic from the source network
VPN traffic can be generated by invoking an application that generates traffic on the VPN tunnel. This can be done from any host in the source network. Enter sufficient wait time.
Wait interval time (in minutes)
1
Continue
Close
Help
Add...
Delete
Seq No
Peers
10.2.2.1
Generate Mirror...


Cisco Configuration Professional
Application
Help
Home
Configure
Monitor
Select Community Member:
192.168.1.1
Configure > Security > VPN > Site-to-Site VPN
VPN
Create Site to Site VPN
Edit Site to Site VPN
Interface Management
Router
Security
Firewall
VPN
VPN Design Guide
Site-to-Site VPN
Easy VPN Remote
Easy VPN Server
Dynamic Multipoint VPN
SSL VPN
GETVPN
VPN Components
Public Key Infrastructure
NAC
Utilities
Flash File Management
Configuration Editor
Save Configuration to PC
Write to Startup Configuration
Telnet
Add...
Delete
Status
Interface
Description
IPSec Policy
Seq No
Peers
Up
Serial3/0
Tunnel to10.2.2.1
SDM_CMAP_1
1
10.2.2.1
Clear Connection
Test Tunnel...
Generate Mirror...